

a Evidence gaps by denominator

His catalytic residue assignable



Asp catalytic residue assignable



Candidates inside a frozen DED alignment



Family resolved to exactly one family



Ledger rows with experimental-negative evidence



High-confidence genomes with local taxonomy



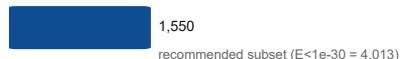
Taxonomy coverage is the last row (higher is better); all other rows are gaps (lower is worse).

b Structural recall path for the deferred layer

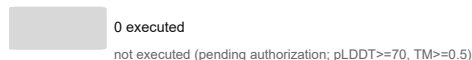
with-lipase deferred layer



strict E-value prefilter ($E < 1e-50$)



structure prediction + Foldseek vs PDB 8YNV



Pending operator authorizations:

[] Fix ESMFold env (openfold + GPU torch)	36 -> 2.2 GPU-h
[] Or approve ColabFold deepfold_v1 / AF2+MSA	needs speed test
[] Approve Stage-2 scale (1,550 sequences)	~36 GPU-h (GPU0)
[] Confirm criteria: TM>=0.5, pLDDT>=70	fail-closed rule
[] Confirm GPU0-only use	GPU1 occupied